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High-intensity interval running is perceived to be more enjoyable than moderate-intensity continuous exercise: Implications for exercise adherence

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Abstract

The aim of this study was to objectively quantify ratings of perceived enjoyment using the Physical Activity Enjoyment Scale following high-intensity interval running versus moderate-intensity continuous running. Eight recreationally active men performed two running protocols consisting of high-intensity interval running (6×3 min at $90\% \dot{V}O_{2\max}$ interspersed with 6×3 min active recovery at $50\% \dot{V}O_{2\max}$ with a 7-min warm-up and cool down at $70\% \dot{V}O_{2\max}$) or 50 min moderate-intensity continuous running at $70\% \dot{V}O_{2\max}$. Ratings of perceived enjoyment after exercise were higher ($P < 0.05$) following interval running compared with continuous running (88 ± 6 vs. 61 ± 12) despite higher ($P < 0.05$) ratings of perceived exertion (14 ± 1 vs. 13 ± 1). There was no difference ($P < 0.05$) in average heart rate (88 ± 3 vs. $87 \pm 3\%$ maximum heart rate), average $\dot{V}O_2$ (71 ± 6 vs. $73 \pm 4\% \dot{V}O_{2\max}$), total $\dot{V}O_2$ (162 ± 16 vs. 166 ± 27 L) or energy expenditure (811 ± 83 vs. 832 ± 136 kcal) between protocols. The greater enjoyment associated with high-intensity interval running may be relevant for improving exercise adherence, since running is a low-cost exercise intervention requiring no exercise equipment and similar relative exercise intensities have previously induced health benefits in patient populations.

Keywords: *Maximal oxygen uptake, intermittent exercise, endurance training*

Introduction

The prevalence of the metabolic syndrome (a cluster of cardiovascular risk factors, including high blood pressure, dyslipidaemia, insulin resistance, and obesity) has grown to epidemic proportions and is reported to have affected 300 million people worldwide (Tjønnå et al., 2008). The proliferation of this epidemic is the result of excess calorie intake and the adoption of a sedentary lifestyle (Hawley & Gibala, 2009). It is increasingly evident that an increase in physical activity is an effective low-cost treatment for the prevention of such symptoms associated with metabolic disorders (Goodyear & Kahn, 1998; Hawley 2004; Hawley & Lessard, 2008; Pedersen & Saltin, 2006; Seals, Hagberg, Hurley, Ehsani, Holloszy, 1984). However, despite the obvious therapeutic potential of exercise interventions, many people are reluctant to engage in exercise, often citing “lack of time” and “lack of enjoyment” as contributing factors (Leslie et al., 1999; Stutts, 2002; Trost, Owen, Bauman, Sallis, & Brown, 2002).

To this end, several investigators (Babraj et al., 2009; Burgomaster, Hughes, Heigenhauser, Bradwell, & Gibala, 2005; Burgomaster et al., 2007, 2008; Gibala et al., 2006; Whyte, Gill, & Cathcart, 2010) have recently adopted short-term high-intensity interval training interventions ($4\text{--}6 \times 30\text{-s}$ maximal cycling Wingate test protocols) as an alternative to more traditional continuous training (40–60 min cycling at $65\% \dot{V}O_{2\max}$) approaches, which are more time-consuming. These authors collectively reported that high-intensity interval training is a time-efficient strategy to induce rapid adaptations in skeletal muscle, as demonstrated by changes in oxidative enzyme protein content/activity as well as improved insulin sensitivity. Although these results have proved insightful, their practical application in improving exercise adherence is limited due to reports of nausea and light-headedness experienced during the exercise protocol (Richards et al., 2010) as well as the requirement for a mechanically braked cycle ergometer, usually found only in the laboratory.

Moreover, it is also difficult to determine if individuals would prefer this type of training intervention due to its short duration *per se* or because they actually perceive the varied activity profile as more *enjoyable* than prolonged approaches, a factor that could potentially improve exercise adherence in the long term.

With this in mind, one potential approach to overcome the barriers associated with lack of exercise may be to develop high-intensity interval *running* interventions that are undertaken at sub-maximal or near maximal intensity. This emphasis upon running is important given that running requires no specialist exercise equipment and because preference of exercise mode is also critical to exercise adherence (Daley & Maynard, 2003; Glass & Chvala, 2001). In this regard, Helgerud et al. (2007) demonstrated that high-intensity interval running (4×4 min at 95% maximum heart rate) is a more efficient training method than moderate-intensity continuous training (45 min at 70% maximum heart rate), as demonstrated by superior improvements in maximum oxygen uptake ($\dot{V}O_{2\max}$), cardiac output, and stroke volume. Such findings are particularly important given that low whole-body $\dot{V}O_{2\max}$ is closely associated with those at an increased risk of metabolic disorders (Mootha et al., 2003). Furthermore, high-intensity interval exercise has also been shown to reduce risk factors associated with the metabolic syndrome compared with moderate-intensity continuous exercise in a variety of patient populations (Haram et al., 2009; Moholdt et al., 2009; Stensvold et al., 2010; Tjønnå et al., 2008, 2009; Wisløff et al., 2007). Interestingly, the exercise intensity associated with high-intensity interval exercise in these populations has been well tolerated despite the low initial fitness of the patients studied. Indeed, Tjønnå et al. (2008) reported informal comments from participants performing high-intensity interval running and found it more motivating to have a varied procedure to follow during each training session, whereas those in the moderate-intensity continuous group found it "quite boring" to exercise continuously for the whole duration. Such comments appear to support the notion that high-intensity interval training is perceived to be more enjoyable than moderate-intensity continuous exercise, although this remains to be tested formally.

The aim of this study, therefore, was to test the hypothesis that high-intensity interval running is perceived to be more enjoyable than moderate-intensity continuous running. To objectively quantify ratings of perceived enjoyment, we employed a previously validated questionnaire (Kenziarski & DeCarlo, 1991), the Physical Activity Enjoyment Scale, which has been shown to be a valid measure to quantify perceived enjoyment and is deemed suitable

for use as a mediator variable in designing interventions to increase physical activity participation (Motl et al., 2001). Furthermore, our particular model of high-intensity interval running has been shown previously in our laboratory to improve $\dot{V}O_{2\max}$ and oxidative adaptations (protein content of COX-IV, PGC-1 α , and SDH activity) of the vastus lateralis and gastrocnemius muscles (Morton et al., 2009).

Methods

Eight healthy recreationally active men volunteered to participate in the study (mean $\pm$ s: age, 25 ± 5 years; weight, 73 ± 5 kg; height, 1.74 ± 0.02 m; body mass index, 24.2 ± 2.2 ; $\dot{V}O_{2\max}$, 57 ± 4 ml $\cdot$ kg $^{-1}$ $\cdot$ min $^{-1}$). All participants undertook exercise two to three times a week and gave written informed consent to participate after details of the study had been explained in full. Participants refrained from additional exercise outside of the study requirements as well as alcohol and caffeine intake for at least 24 h prior to any of the test sessions. The study received approval from the Ethics Committee of Liverpool John Moores University.

All participants were initially assessed for $\dot{V}O_{2\max}$ using an incremental exercise test performed on a motorized treadmill (HP Cosmos, Germany). Oxygen uptake was measured continuously during exercise using an on-line gas analysis system (Metamax, Cortex). The test began with a 2-min stage at a treadmill speed of 10 km $\cdot$ h $^{-1}$ followed by 2-min stages at 12 km $\cdot$ h $^{-1}$, 14 km $\cdot$ h $^{-1}$, and 16 km $\cdot$ h $^{-1}$. On completion of the 16 km $\cdot$ h $^{-1}$ stage, the treadmill was inclined by 2% every 2 min until volitional exhaustion despite strong verbal encouragement. Maximum oxygen uptake was taken as the highest $\dot{V}O_2$ value obtained in any 10-s period and was taken as having been achieved when meeting the following end-point criteria in accordance with the guidelines of the British Association of Sport and Exercise Sciences: (1) heart rate within 10 beats $\cdot$ min $^{-1}$ of age-predicted maximum ($220 - \text{age}$); (2) respiratory exchange ratio > 1.15 ; and (3) plateau of $\dot{V}O_2$ despite increased workload (Bird & Davison, 1997). At least 48 h after assessment of $\dot{V}O_{2\max}$, participants undertook a running test that consisted of similar time points at each stage (3 min) as the high-intensity interval running protocol so as to provide a clearer profile of $\dot{V}O_2$, which was used to accurately match the participants' running velocities with their respective exercise intensities. The exercise protocol commenced at a treadmill speed of 8 km $\cdot$ h $^{-1}$ and was increased by 1 km $\cdot$ h $^{-1}$ every 3 min until at least 95% $\dot{V}O_{2\max}$ had been reached (taken from previous $\dot{V}O_{2\max}$ test). Expired gases were collected using the same on-line gas analysis system (Metamax, Cortex).

At least 5 days after the assessment of $\dot{V}O_{2\max}$ and running economy, participants completed two running exercise protocols consisting of high-intensity interval running and moderate-intensity continuous running in a fully randomized and crossover design (Altman, 1991). The exercise protocols were performed 7 days apart after an overnight fast and at the same time of day to eliminate any circadian effects on the physiological responses to the exercise protocols (Reilly & Brooks, 1986). Both protocols were performed on the same motorized treadmill (HP Cosmos, Germany). The high-intensity interval running protocol commenced with a 7-min warm-up at a running velocity corresponding to 70% $\dot{V}O_{2\max}$ followed by six 3-min bouts at a running velocity corresponding to 90% $\dot{V}O_{2\max}$. The high-intensity intervals were separated by 3-min active recovery periods at a running velocity corresponding to 50% $\dot{V}O_{2\max}$. Following the interval and recovery periods, participants then performed a 7-min cool down at a running velocity corresponding to 70% $\dot{V}O_{2\max}$. The exercise protocol therefore consisted of 18 min of high-intensity exercise and 18 min of active recovery time, giving a total interval exercise time of 36 min. When including the warm-up and cool-down, the total duration of the exercise protocol was 50 min. The average intensity during the high-intensity interval running protocol, when quantified according to average running velocity, equated to 70% $\dot{V}O_{2\max}$. The moderate-intensity continuous running protocol consisted of 50 min continuous running at a running velocity corresponding to 70% $\dot{V}O_{2\max}$. In this way, the two exercise protocols were matched for exercise duration (50 min), average

intensity (70% $\dot{V}O_{2\max}$), and distance run (9843 ± 494 m). Given that there is no applied external load with horizontal motorized treadmill running at 0% incline (as is the case in conventional ergometry), current biomechanical methods are not well enough established to accurately measure work done. Nevertheless, in this type of activity, work done is best quantified by the total energy cost of each protocol, as assessed by total oxygen consumption (which was also matched in both exercise protocols).

Oxygen uptake (Metamax, Cortex) and heart rate (Polar S610i, Kempele, Finland) were measured continuously during each exercise protocol and participants also reported ratings of perceived exertion (RPE) at 1-min intervals during exercise (Borg, 1970). Energy expenditure was estimated during exercise via indirect calorimetry assuming a non-protein respiratory exchange ratio (Zuntz, 1901). Immediately upon completion of each protocol, participants also reported ratings of perceived enjoyment according to the Physical Activity Enjoyment Scale validated by Kendzierski and DeCarlo (1991). Each item was rated on a 7-point bipolar scale with 4 representing a neutral point in terms of how much the respondent enjoyed the exercise (Table I).

Statistical analysis was conducted using the Statistical Package for Social Sciences software program (version 15). All data were initially analysed for normality according to the Shapiro-Wilks test. Once verified, a comparison of the average physiological responses between exercise protocols was assessed using Student's *t*-tests for paired samples, and a two-way general linear

Table I. A reproduction of the Physical Activity Enjoyment Scale questionnaire demonstrating the 7-point bipolar scale.

* I enjoy it	1	2	3	4	5	6	7	I hate it
I feel bored	1	2	3	4	5	6	7	I feel interested
I dislike it	1	2	3	4	5	6	7	I like it
* I find it pleasurable	1	2	3	4	5	6	7	I don't find it pleasurable
* I am very absorbed in this activity	1	2	3	4	5	6	7	I am not at all absorbed in this activity
It's no fun at all	1	2	3	4	5	6	7	It's a lot of fun
* I find it energizing	1	2	3	4	5	6	7	I find it tiring
It makes me depressed	1	2	3	4	5	6	7	It makes me happy
* It's very pleasant	1	2	3	4	5	6	7	It's very unpleasant
* I feel good physically while doing it	1	2	3	4	5	6	7	I feel bad physically while doing it
* It's very invigorating	1	2	3	4	5	6	7	It's not at all invigorating
I am very frustrated by it	1	2	3	4	5	6	7	I am not at all frustrated by it
* It's very gratifying	1	2	3	4	5	6	7	It's not at all gratifying
* It's very exhilarating	1	2	3	4	5	6	7	It's not at all exhilarating
It's not at all stimulating	1	2	3	4	5	6	7	It's very stimulating
* It gives me a strong sense of accomplishment	1	2	3	4	5	6	7	It doesn't give me a strong sense of accomplishment
* It's very refreshing	1	2	3	4	5	6	7	It's not at all refreshing
I felt as though I would rather be doing something else	1	2	3	4	5	6	7	I felt as though there is nothing else I would rather be doing

* Denotes reversal when scoring.

model was used to determine whether there was an interaction or any significant differences over time and between conditions for heart rate and RPE. A Pearson's rank correlation was run to assess whether a relationship existed between perceived enjoyment and $\dot{V}O_{2\max}$. All data in the text, figures, and tables are presented as means and standard deviations (s) with P -values ≤ 0.05 indicating statistical significance.

Results

Psycho-perceptual responses to high-intensity interval and moderate-intensity continuous protocols

Ratings of perceived enjoyment were significantly higher ($P=0.004$) in the high-intensity interval running condition compared with the moderate-intensity continuous running condition (Figure 1A). In

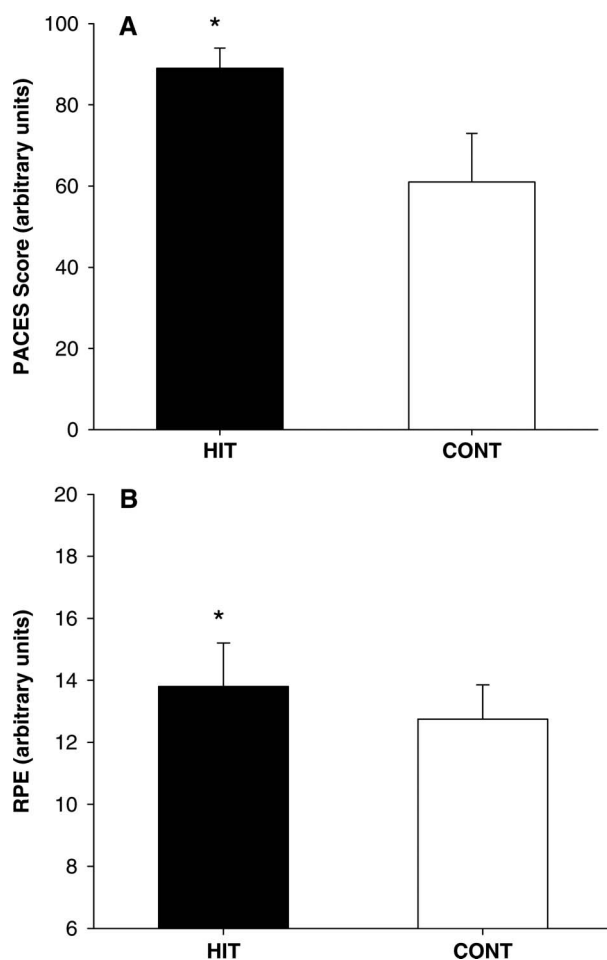


Figure 1. (A) Ratings of perceived enjoyment (PACES score) after completing the high-intensity interval (HIT) protocol and moderate-intensity continuous (CONT) protocol, as quantified using the Physical Activity Enjoyment Scale (PACES). (B) Average ratings of perceived exertion (RPE), as quantified using the Borg scale. *Significantly different from CONT protocol ($P < 0.05$).

spite of the higher enjoyment during the high-intensity interval running protocol, average ratings of perceived exertion were also significantly higher ($P=0.015$) during high-intensity interval running compared with the moderate-intensity continuous running protocol (Figure 1B). In addition, when comparing RPE during exercise between conditions there was a significant interaction ($P=0.000$), main effect of time ($P=0.000$), and main effect of condition ($P=0.027$) for the high-intensity interval running and moderate-intensity continuous running protocols (Figure 2A).

Physiological responses to high-intensity interval and moderate-intensity continuous protocols

There was no significant difference in average heart rate ($P=0.547$), average percent maximum heart rate ($P=0.528$), total $\dot{V}O_2$ ($P=0.358$), average $\dot{V}O_2$ ($P=0.389$), average $\% \dot{V}O_{2\max}$ ($P=0.305$) or energy expenditure ($P=0.383$) between the high-intensity interval running and moderate-intensity continuous running protocols (Table II). However,

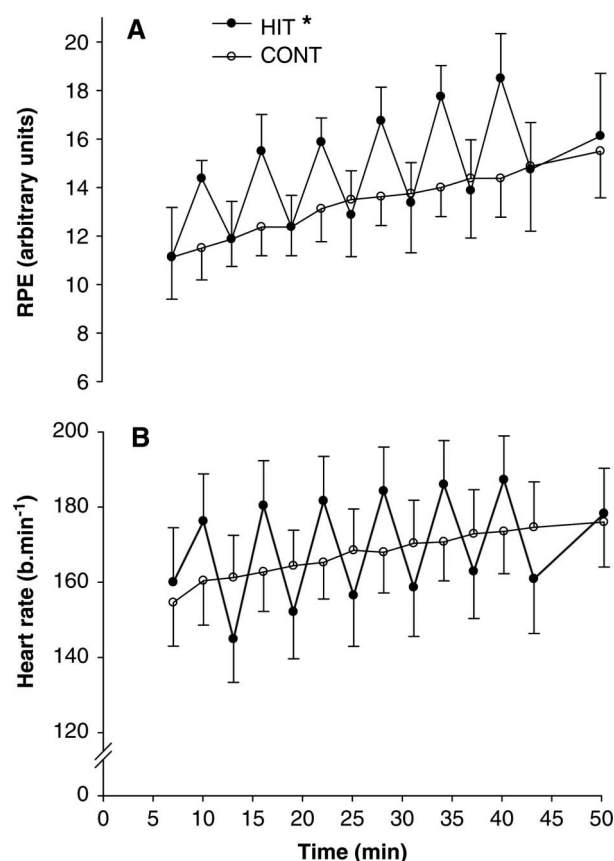


Figure 2. (A) Ratings of perceived exertion (RPE) and (B) heart rates after each stage during the high-intensity interval (HIT) protocol and moderate-intensity continuous (CONT) protocol. *Significant main effect of time, condition and interaction ($P < 0.05$).

Table II. A comparison of the average physiological responses induced by the two running protocols (mean $\pm$ s).

	Protocol	
	High-intensity interval protocol	Moderate-intensity continuous protocol
Heart rate (beats $\cdot$ min ⁻¹)	167 $\pm$ 12	165 $\pm$ 9
Heart rate (%HR _{max})	88 $\pm$ 3	87 $\pm$ 3
Total $\dot{V}O_2$ (L)	162 $\pm$ 16	166 $\pm$ 27
Average $\dot{V}O_2$ (L $\cdot$ min ⁻¹)	3.23 $\pm$ 0.32	3.32 $\pm$ 0.53
Average $\dot{V}O_2$ (% $\dot{V}O_{2max}$)	71 $\pm$ 6	73 $\pm$ 4
Energy expenditure (kcal)	811 $\pm$ 83	832 $\pm$ 136

Note: HR_{max} = maximum heart rate.

there was a significant interaction and main effect for time on both heart rate and $\dot{V}O_2$ when comparing these variables between protocols throughout the 50 min of exercise ($P=0.000$ for all effects; heart rate data shown in Figure 2B), thus highlighting the obvious difference in activity profile between exercise protocols.

Discussion

The aim of the present study was to quantify subjective feelings of enjoyment generated by undertaking an acute bout of high-intensity interval running and moderate-intensity continuous running. Our data demonstrate that recreationally active males perceive high-intensity interval running to be more enjoyable than moderate-intensity continuous running, despite higher ratings of RPE during interval exercise. In addition, this experimental approach represents a methodological development, as it is the first study to match high-intensity interval running and moderate-intensity continuous running for duration, distance run, average intensity, and total oxygen consumption and energy expenditure (i.e. work done).

We quantified ratings of perceived enjoyment according to the Physical Activity Enjoyment Scale and observed higher scores following high-intensity interval running compared with moderate-intensity continuous running. We acknowledge that our participants were recreationally active ($\dot{V}O_{2max}$: 57 ml $\cdot$ kg⁻¹ $\cdot$ min⁻¹) and that the perception of effort/enjoyment experienced by these individuals may be different from that of individuals who do not regularly engage in physical activity. Nevertheless, our results are in agreement with subjective reports from previous authors examining cardiovascular diseased ($\dot{V}O_{2max}$: 13 ml $\cdot$ kg⁻¹ $\cdot$ min⁻¹) and metabolic syndrome patients ($\dot{V}O_{2max}$: 32–36 ml $\cdot$ kg⁻¹ $\cdot$ min⁻¹) who reported informal comments of increased enjoyment during high-intensity

interval running compared with moderate-intensity continuous running (Tjønnå et al., 2008; Wisløff et al., 2007). These authors also observed that high-intensity interval exercise induced greater improvements in endothelial function and aerobic capacity, and improved insulin signalling in skeletal muscle compared with moderate-intensity continuous exercise training. While we cannot comment on the health-promoting effects of our specific exercise protocol, both of the aforementioned investigations used similar relative training intensities (4 $\times$ 4-min intervals at 95% maximum heart rate) to those studied here, thus highlighting the potential for our exercise model to induce health benefits. Furthermore, we have previously observed that our model of high-intensity interval running increases protein content of PGC-1 α of the vastus lateralis and gastrocnemius muscles (Morton et al., 2009) within the physiological range associated with improved insulin sensitivity in humans (Lira, Benton, Yan, & Bonen, 2010).

Comments from the patients studied by Wisløff et al. (2007) and Tjønnå et al. (2008) suggested that greater post-exercise feelings of enjoyment are due to the varied nature of the activity profile inherent to high-intensity training compared with the “boring” steady-state continuous approach, a view also reported by the current participants. It is presently difficult to therefore distinguish if enhanced feelings of enjoyment associated with high-intensity interval running are specifically due to the varied activity, periods of increased exercise intensity or a combination of both factors. In this regard, it is possible that enjoyment may also be enhanced with varied activity profiles that are undertaken at lower intensities, although the clinical relevance of such protocols are questionable given that exercise intensity appears to be a key determinant of physiological adaptations (Egan et al., 2010).

In spite of higher ratings of enjoyment associated with high-intensity interval running, ratings of perceived exertion were significantly higher following high-intensity interval running than moderate-intensity continuous running. A positive relationship between higher ratings of enjoyment and higher ratings of perceived exertion has been demonstrated previously in recreationally active participants (Rac-deke, 2007), suggesting that individuals who engage in regular exercise and thus are aerobically fitter, feel a greater sense of challenge, stimulation or accomplishment after training sessions that are perceived to be more demanding. However, Pearson’s correlation test demonstrated no significant relationship between $\dot{V}O_{2max}$ and Physical Activity Enjoyment Scale scores during high-intensity interval running ($P=0.823$) or moderate-intensity continuous running ($P=0.977$), suggesting that initial fitness does

not seem to be a major determinant of exercise enjoyment. In keeping with this thesis, it might be surmised that the varied activity profiles inherent to the Wingate cycling training protocols may also result in enhanced enjoyment (Babraj et al., 2009; Burgomaster et al., 2005, 2007, 2008; Gibala et al., 2006; Whyte et al., 2010), although unfortunately these studies did not address or comment on this issue. However, the 30-s maximal efforts inherent to these protocols are associated with intense feelings of nausea and light-headedness (Richards et al., 2010). In contrast, in relation to the participants studied in our laboratory both here and previously (Morton et al., 2009), as well as the patient populations studied by other groups (Tjønnå et al., 2008, 2009; Wisløff et al., 2007), there are no reports of ill-feeling during or after the high-intensity interval running protocols undertaken at near maximal or sub-maximal intensities.

To isolate the effects of “exercise activity profile” on enjoyment as opposed to duration *per se*, we evaluated the perceived enjoyment responses after two exercise protocols that were matched for duration, distance run, average intensity, oxygen consumption, and energy expenditure. To the authors’ knowledge, this experimental approach is unique, as it is the first study to match high-intensity interval and moderate-intensity continuous running protocols for all the aforementioned indices. Indeed, when prescribing and quantifying exercise intensity during high-intensity interval running and moderate-intensity continuous running protocols on the basis of percent maximum heart rate, Helgerud et al. (2007) and Tjønnå et al. (2008) matched their interventions for energy expenditure but not duration, distance run or average intensity. While this approach of prescribing, quantifying, and monitoring exercise intensity on the basis of heart rate measurements is practically advantageous it is not without limitations, as heart rate *per se* does not always correspond to energy production (Gilman, 1996). Furthermore, to maintain exercise intensity in the target heart rate zones, these authors had to reduce the running velocity during the training session so as to offset the effects of cardiovascular drift. However, in this way, the absolute workload is reduced, therefore lowering the overall training stress. In an effort to provide a more controlled exercise stress, we chose to prescribe exercise intensity on the basis of running velocity corresponding to a particular $\% \dot{V}O_{2\max}$, whereby the absolute workload and degree of metabolic signalling within the muscle was not likely to be compromised during exercise within or between individuals (Morton, 2007). It should be noted, however, that the matched physiological responses observed here are likely to be highly specific to our chosen exercise intensities and work-

to-rest ratios (O’Brien, Wibskov, Knez, Paton, & Harvey, 2008).

Conclusions

In conclusion, the present results demonstrate that ratings of perceived enjoyment are greater after undertaking an acute bout of high-intensity interval running compared with moderate-intensity continuous running. Our results are of practical relevance given that running is a low-cost exercise intervention requiring no specialist exercise equipment and that similar relative exercise intensities have been shown to improve cardiovascular and metabolic health in clinical populations. High-intensity interval running may therefore prove an effective strategy to increase long-term exercise participation and improve human health and future studies (longitudinal in design) are required to test this hypothesis.

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